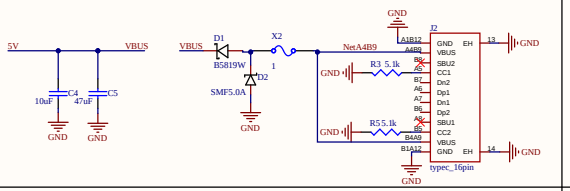
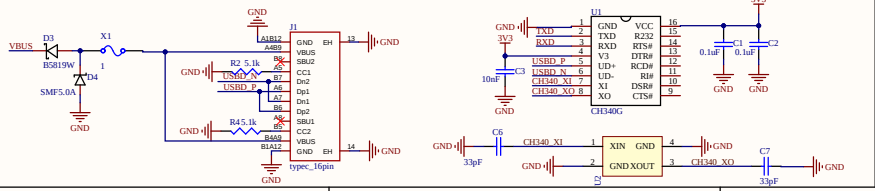
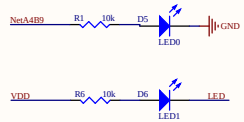
USB仅供电



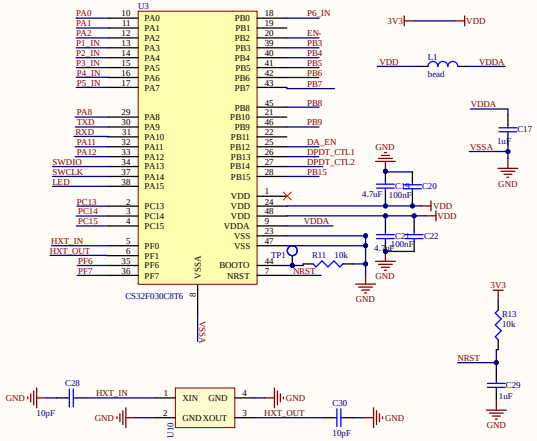
USB转串口



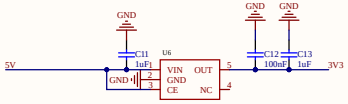
电源指示灯



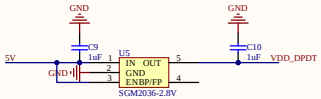
控制核心



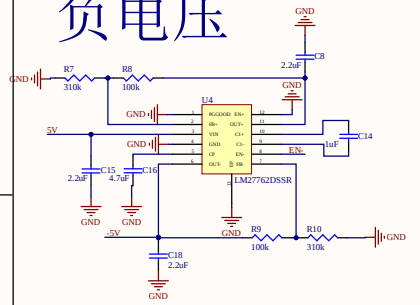
5v转3.3v



5v转2.8v



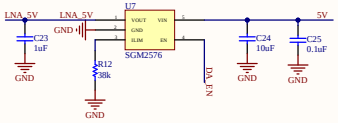
负电压



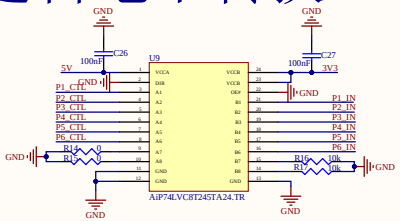
SWD下载口



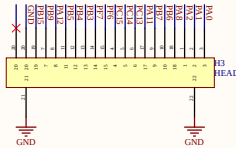
DA使能开关



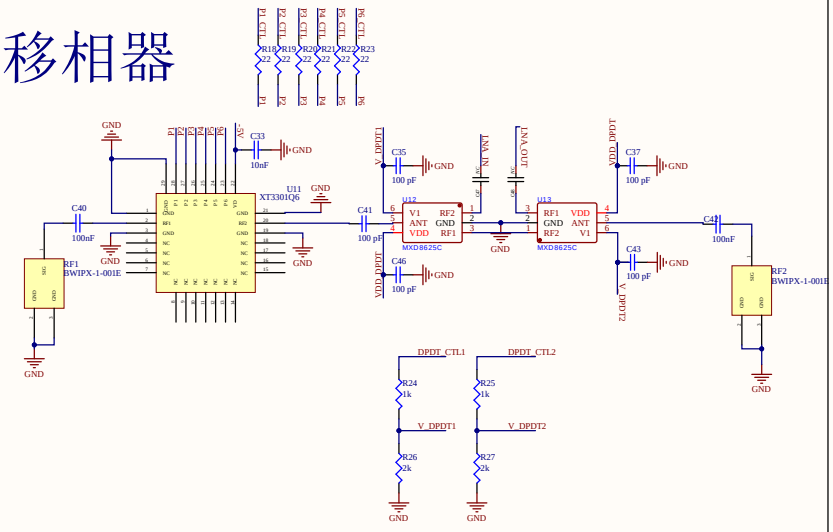
逻辑电平转换



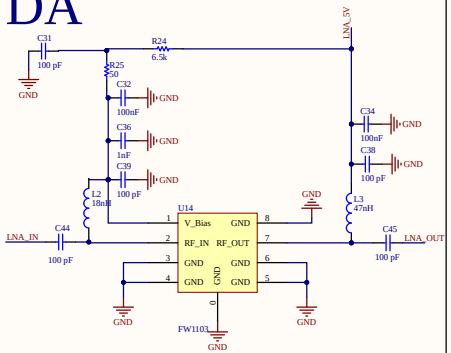
IO口



移相器



DA



Title		
Size: A2	Number	Revision
Date: 10/17/2026		Sheet of
File: C:\Users\...\SHIFTER.SchDoc		Drawn By:

Comment	Description	Designator	Footprint	LibRef	Quantity
0.1uF	0402, Capacitor	C1, C2, C25	0402_Z	CAP	3
10nF	0402	C3, C33	0402_Z	CAP	2
10uF	0402, Capacitor	C4, C24	0603_Z	CAP	2
47uF	0402	C5	0805	CAP	1
33pF	0402	C6, C7	0402_Z	CAP	2
2.2uF	0402	C8, C15, C18	0402_Z	CAP	3
1uF	Capacitor	C9, C10, C11, C13, C23	0603_Z	Cap	5
100nF	Capacitor, 0402	C12, C40, C42	0603_Z	Cap	3
1uF	0402	C14, C17, C29	0402_Z	CAP	3
4.7uF	0402	C16, C19, C21	0603_Z	CAP	3
100nF	0402, Capacitor	C20, C22, C26, C27, C32, C34	0402_Z	CAP	6
10pF	0402	C28, C30	0402_Z	CAP	2
100 pF	Capacitor	C31, C35, C37, C38, C39, C41, C43, C44, C45, C46	0402_Z	Cap	10
1nF	Capacitor	C36	0402_Z	Cap	1
B5819W	肖特基二极管	D1, D3	BW5819W	B5819W	2
SMF5.0A		D2, D4	SMF5.0A	SMF5.0A	2
LED0	Typical INFRARED GaAs LED	D5	0603_Z	LED0	1
LED1	Typical INFRARED GaAs LED	D6	0603_Z	LED0	1
HEADER20		H3	Header20	HEADER20	1
typec_16pin		J1, J2	TYPEC_16PIN	typec_16pin	2
bead	INDUCTOR	L1	0603_Z	INDUCTOR1_1	1
18nH	Inductor	L2	0402_Z	Inductor	1
47nH	Inductor	L3	0402_Z	Inductor	1
10k	Resistor	R1, R6, R11, R13, R16, R17	0402_Z	Res3	6
5.1k	Resistor	R2, R3, R4, R5	0402_Z	Res3	4
310k	Resistor	R7, R10	0402_Z	Res3	2
100k	Resistor	R8, R9	0402_Z	Res3	2
38k	Resistor	R12	0402_Z	Res1	1
0	Resistor	R14, R15	0402_Z	Res3	2
22	Resistor	R18, R19, R20, R21, R22, R23	0402_Z	Res3	6
6.5k	Resistor	R24	0402_Z	Res1	1
1k	Resistor	R24, R25	0402_Z	Res3	2
50	Resistor	R25	0402_Z	Res1	1
2k	Resistor	R26, R27	0402_Z	Res3	2
BWIPX-1-001E		RF1, RF2	SMA-SMD_BWSMA-KE-P001	BWIPX-1-001E	2
TP_P		TP1	TP_0.8	TP_P	1
CH340G		U1	CH340G	CH340G	1
12MHz		U2	CRYSTAL-SMD_4P-L3.2-W2.5-BL	XTAL 3225	1
CS32F030C8T6		U3	LQPF48	CS32F030C8T6	1
LM27762DSSR		U4	LM27762	LM27762DSSR	1
SGM2036-2.8V		U5	SOT-23-5	sgm2036	1
SGM2036-3.3YN5G/TR	0.8V-5V可调输出线性稳压器	U6	SOT-23-5	LDO_ADJ_SGM2036	1
SGM2576		U7	SGM2576	SGM2576	1
HDGC1251WR-4P	复旦微单片机程序下载接口	U8	J4_1.25	MX1.25_弯针	1
AiP74LVC8T245TA24.TR	具有三态输出的8位双电源转换收发器，可实现双向电平转换	U9	AIP74LVC8T245TA24.TR	AiP74LVC8T245TA24.TR	1
8MHz		U10	CRYSTAL-SMD_4P-L3.2-W2.5-BL	XTAL 3225	1
XT3301Q6		U11	XT3301Q6	XT3301Q6	1
MXD8625C		U12, U13	MXD8625C	MXD8625C	2
FW1103		U14	FW1103W	FW1103	1
Fuse	Fuse	X1, X2	F1206	Fuse	2